Total number of printed pages = 5

(Sem-2) MAT-~~HC~~-COR-2-01.

2024

MATHEMATICS

~~(HONOURS)~~ (MAJOR).

Paper : 2-01

(Differential Equations)

Full Marks : 60

Time : 3 hours

The figures in the margin indicate full marks for the questions.

1. Answer ***all*** the questions : $1\times7=7$

   (a) Give example of a linear differential equation.

   (b) Is the differential equation $\frac{dy}{dx}=\frac{xy}{x+y}$ homogenous ?

   (c) Write the general solution of differential equation

   $y = px + p^2$, where $p = dy / dx$.

Contd.

(d) Under what condition differential equation $Mdx + Ndy = 0$ is exact with usual meaning of symbols ?

(e) What is radioactive substance ?

(f) Write the order of differential equation

$$\sqrt{\frac{d^3y}{dx^3} + 1} = \frac{dy}{dx}$$

(g) Give the complementary function of $(D^2 - 3D + 2)y = 5e^x$.

2. Answer ***any four*** questions : 2×4=8

(a) Show that differential equation of all circles touching x axis at the origin is

$$(x^2 - y^2)dy - 2xydx = 0$$

(b) Solve : $(a + x^2)dy + (1 + y^2)dx = 0$

(c) Show that functions $y_1 = \sin 2x, y_2 = \cos 2x$ are linearly independent.

(d) Define mathematical modelling. Give two main objectives of modelling.

(e) Define compartmental model with example.

3. Answer ***any three*** questions : 5×3=15

(a) Solve : $\frac{dy}{dx} = \sec(x+y)$

(b) Solve : $p^2 - p(e^x + e^{-x}) + 1 = 0$

(c) Solve : $(D^2 + 4)y = x^2$

(d) Find particular integral of $y'' + 3y' + 4y = 3x + 2$ by method of undetermined coefficients.

(e) For exponential growth model, find an expression for time for the population to double in size.

4. Answer ***any three*** questions : 10×3=30

(a) (i) Solve by the method of variation of parameters

$$\frac{d^2y}{dx^2} + n^2y = \sec nx$$ 6

(ii) Solve : $y'' + 4y = x\cos x$ 4

(b) (i) Solve : $x^3y''' - x^2y'' + 2xy' - 2y = x^2$ 5

(ii) Solve : $y = (1+p)x + ap^2$ 5

(c) (i) Solve : $\frac{dy}{dx} + \frac{y}{x} = \frac{y^2}{x^2}$ 5

(ii) Solve : $\frac{dy}{dx} + \frac{1-2x}{x^2}y = 1$ 5

(d) (i) Solve : $\frac{dy}{dx} = \frac{6x-2y-7}{2x+3y-6}$ 6

(ii) Solve : $(xy^2 + 3e^{x^{-3}})dx - x^2y\,dy = 0$ 4

(e) (i) Solve :

$$(x+a)^2\frac{d^2y}{dx^2} - 4(x+a)\frac{dy}{dx} + by = x$$ 6

(ii) Show that $e^x \sin x$ and $e^x \cos x$ are linearly independent solution of $y'' - 2y' + 2y = 0$ and find the general solution $y(x)$ with initial conditions $y(0) = 2, y'(0) = 3$. 4

(f) Derive the mathematical model for density-dependent growth of population. Hence derive the logistic equation. Thereafter interpret the population growth corresponding to carrying capacity of the environment.
